

Performance Testing & Enterprise Benchmarking

Multi-Stack Evaluation: Salesforce Bulk API, Tableau & Power BI Analytics, GenAI Models & UAT Sign-Off

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Project Domain	Salesforce, Tableau, GenAI, Power BI & AI/ML	Assessment Standard	Enterprise Benchmarking & UAT

1. Executive Summary & Multi-Stack Environment

This technical evaluation assesses data throughput, transformation latency, model inference response times and dashboard render stability across interconnected enterprise platforms: Salesforce, Generative AI endpoints, Power BI, Tableau and Machine Learning pipelines.

Platform	Scope of Evaluation
Salesforce Cloud	Bulk API v2.0 ingestion into Account, Contact and Opportunity objects with custom Apex triggers.
GenAI / LLM Stack	Generative AI inference pipeline evaluated for token velocity, time-to-first-token (TTFT) and context window scaling under concurrent prompt requests.
Power BI & Tableau Server	DirectQuery and Hyper in-memory extract refreshes supporting executive analytics, Level of Detail (LOD) calculations and complex DAX visual queries.
AI / ML Pipelines	Batch feature store extraction and distributed Spark model inference engines for predictive business scoring.
User Acceptance Testing (UAT)	End-to-end business process verification ensuring complete functional alignment across operational departments.

2. Workload Profiles & Test Scenarios

Scenario ID	Target Technology	Workload Volume (N)	Execution Mode	Primary SLA Metric
TS-SF-01	Salesforce Data Import	50,000 records	Bulk API v2.0 Parallel	Ingestion speed $\geq$ 150 rec/sec
TS-AI-02	GenAI Performance	1,000 prompt queries	Concurrent LLM Inference	Token velocity $\geq$ 45 tokens/sec
TS-BI-03	Power BI Refresh	5,000,000 rows	Incremental DirectQuery	Refresh time $\leq$ 120 seconds
TS-TB-04	Tableau Analytics	2,500,000 rows	Hyper Extract Engine	Visual render $\leq$ 2.5 seconds
TS-ML-05	ML Pipeline	100,000 feature vectors	Distributed Spark ML	Batch duration $\leq$ 180 seconds
TS-UAT-06	User Acceptance (UAT)	150 end-to-end critical flows	Functional Verification	Pass rate = 100%

3. Empirical Performance Benchmarking Results

Test ID	Platform / Stack	Duration (t)	Throughput (T)	Avg Latency	Status
TS-SF-01	Salesforce Bulk API	210.0 sec	238.1 rec/sec	4.20 ms/rec	PASSED
TS-AI-02	GenAI Inference API	18.2 sec/query	52.4 tokens/sec	19.08 ms/token	OPTIMIZED
TS-BI-03	Power BI Dataset	84.5 sec	59,171 rows/sec	0.017 ms/row	PASSED
TS-TB-04	Tableau Hyper Extract	1.85 sec (render)	1,351,351 rows/sec	0.74 µs/row	PASSED
TS-ML-05	ML Batch Pipeline	112.0 sec	892.8 vectors/sec	1.12 ms/vector	PASSED

SLA Compliance Margins

Test ID	SLA Target	Measured	Margin vs SLA
TS-SF-01	≥ 150 rec/sec	238.1 rec/sec	+58.7% above target
TS-AI-02	≥ 45 tokens/sec	52.4 tokens/sec	+16.4% above target
TS-BI-03	≤ 120 sec	84.5 sec	35.5 sec (29.6%) faster
TS-TB-04	≤ 2.5 sec	1.85 sec	0.65 sec (26.0%) faster
TS-ML-05	≤ 180 sec	112.0 sec	68 sec (37.8%) faster

All five performance scenarios met their SLA targets with positive headroom.

4. Mathematical Formulas & Throughput Calculations

1. Record / Data Throughput Rate (T)

Quantifies the rate of record ingestion or feature vectorization per unit of time.

$$T = \frac{N}{t} = \frac{50,000 \text{ records}}{210.0 \text{ seconds}} \approx \textbf{238.10 records/second}$$

2. GenAI Token Generation Throughput (T_{token})

Measures the streaming throughput of LLM output generation across concurrent client sessions.

$$T_{\text{token}} = \frac{\sum_{i=1}^k \text{Tokens Output}_i}{t_{\text{generation}}} = \frac{953.68 \text{ tokens}}{18.2 \text{ seconds}} \approx \textbf{52.40 tokens/second}$$

3. Optimization Efficiency Gain (η)

Evaluates processing efficiency improvements achieved through index creation, query refactoring and caching.

$$\eta = \frac{t_{\text{unoptimized}} - t_{\text{optimized}}}{t_{\text{unoptimized}}} \times 100\% = \frac{700 \text{ s} - 210 \text{ s}}{700 \text{ s}} \times 100\% = \textbf{70.0\%}$$

5. Bottleneck Diagnostics & System Triage

Platform	Identified Bottleneck & Cause	Implemented Optimization
Salesforce Cloud	Governor limits hit during Apex triggers on bulk updates.	Bypassed non-critical triggers; switched to Bulk API v2.0 parallel ingestion.
GenAI / LLM	High TTFT caused by large context window parsing.	Implemented KV-cache reuse, prompt compression and dynamic GPU batching.
Power BI	Slow DirectQuery visuals due to unindexed RLS rules.	Refactored DAX measures, added composite models, materialized summary tables.
Tableau Server	Latency on complex Level of Detail (LOD) expressions.	Pre-computed LODs in Hyper extracts; scheduled background pre-warming.
Machine Learning	I/O throttling during high-dimensional feature retrieval.	Migrated active feature store tables to an in-memory Redis cluster with async fetch.

6. User Acceptance Testing (UAT) Sign-Off Matrix

UAT Module	Acceptance Criteria	Stakeholder Feedback	Sign-Off Status
Salesforce Sync	Bulk records match legacy sources with zero orphan records.	Verified and approved by CRM Ops.	PASSED / SIGNED
GenAI Functional	Output complies with corporate safety and formatting rules.	Approved by Governance Team.	PASSED / SIGNED
Power BI / Tableau	Visual filtering renders within 2.5 s with exact fidelity.	Key KPI metrics verified by Executives.	PASSED / SIGNED
AI / ML Scoring	Prediction pipeline delivers precision and recall > 92%.	Validated by Data Science Team.	PASSED / SIGNED

Final Conclusion & Sign-Off Recommendation

The performance benchmarking and User Acceptance Testing (UAT) across Salesforce, GenAI, Power BI, Tableau and Machine Learning pipelines confirm that all evaluated modules meet enterprise reliability and SLA criteria. With an overall processing efficiency gain of $\eta = 70.0\%$, the integrated system is **officially approved for production deployment**.